Boundary-Preserved Deterministic QEM

Out-of-Core Simplification for Large Triangle Meshes

Publisher: Make Your Own World
Version: 1.0

Abstract

Large triangle meshes can exceed practical working-memory limits even when their binary STL files fit on disk. This report presents a bounded-memory pipeline that partitions a binary STL into deterministic adaptive octree tiles, simplifies tiles concurrently, freezes partition boundaries, checkpoints every accepted result, and streams a final STL without constructing a global vertex or face array.

On a 4,126,315-triangle reference mesh, eight CPU workers completed the tiled reduction in 4.67 seconds compared with 27.34 seconds for one worker, a 5.85x speedup. Sequential, parallel, and cached runs produced byte-identical 249,679-triangle outputs. The requested count was 249,184. Boundary preservation retained 495 triangles. Connected-surface tests measured zero assembly loss and no new open partition seams.

Contributions

The contribution is the integrated production contract: singular triangle ownership, stable allocation, immutable float32 partition coordinates, memory-bounded scheduling, atomic content-validated checkpoints, cancellation-safe streamed assembly, and byte-level determinism. Octrees, Morton order, QEM, locked boundaries, and external-memory processing are established foundations and are attributed accordingly.

1. The coupled problem

A compact triangle stream expands during reduction into welded vertices, indexed faces, adjacency, quadrics, candidate contractions, and temporary native state. Peak memory is therefore the sum of simultaneous structures, not the input file size:

$$M_{\text{peak}} \approx M_{\text{source}} + M_{\text{index}} + M_{\text{adjacency}} + M_{\text{quadrics}} + M_{\text{candidates}} + M_{\text{temporary}}$$

$\mathbf{M_{peak}}$ is peak resident geometry memory; $\mathbf{M_{source}}$ is the streamed source state; $\mathbf{M_{index}}$ is the spatial index; $\mathbf{M_{adjacency}}$ is connectivity state; $\mathbf{M_{quadrics}}$ stores error matrices; $\mathbf{M_{candidates}}$ stores proposed collapses; and $\mathbf{M_{temporary}}$ is native working memory. All terms use bytes.

M_peak is peak resident geometry memory. Every term is live during at least part of an in-memory reduction. Streaming the reader alone does not solve the problem when topology and contraction state still scale with the complete mesh.

Partitioning introduces six linked challenges: one owner for every source triangle, continuous seams, a deterministic global target, bounded concurrency, validated recovery, and stable output order. Fixing one while ignoring the others produces a reducer that is fast in a narrow test but unsafe for long production runs.

Boundary-Preserved Deterministic QEM pipeline



2. Local QEM mathematics

For plane p = (a,b,c,d)^T and homogeneous vertex v_bar = (x,y,z,1)^T, the plane quadric and accumulated vertex quadric are:

$$Q_p = pp^T, \qquad Q_v = \sum_{p \in P(v)} Q_p$$

p is a normalized incident face plane in homogeneous form; $\mathbf{Q_p}$ is that plane's 4-by-4 error matrix; **P(v)** is the set of planes incident on vertex **v**; and $\mathbf{Q_v}$ is the accumulated vertex quadric.

The contraction cost for vertices v1 and v2 at candidate v'_bar is:

$$E(\bar{\mathbf{v}}') = \bar{\mathbf{v}}'^T (Q_{v_1} + Q_{v_2}) \bar{\mathbf{v}}'$$

$\mathbf{v}'$ is the proposed homogeneous replacement vertex; $\mathbf{Q_{v1}}$ and $\mathbf{Q_{v2}}$ are the endpoint quadrics; and **E** is the estimated squared plane-distance error used to rank the contraction.

p is an incident face plane. P(v) is the set of planes incident on v. Q_v accumulates squared distance penalties. E is the estimated geometric error introduced by the candidate contraction. The native local reducer performs these contractions. The method described here supplies safe ownership, constraints, scheduling, recovery, and assembly around it.

3. Spatial ownership

For triangle f with vertices x_f1, x_f2, and x_f3, its persistent owner is selected from its centroid:

$$\mathbf{c}_f = \frac{\mathbf{x}_{f,1} + \mathbf{x}_{f,2} + \mathbf{x}_{f,3}}{3}$$

f is one source triangle; $\mathbf{x_{f,1}}$, $\mathbf{x_{f,2}}$, and $\mathbf{x_{f,3}}$ are its vertex positions; and $\mathbf{c_f}$ is its centroid in mesh coordinates.

For lower bound b_min, extent d, level l, grid width g = 2^l, and small epsilon:

$$\mathbf{u}_f = \mathrm{clip}\left(\left\lfloor g \frac{\mathbf{c}_f - \mathbf{b}_{\min}}{\max(\mathbf{d}, \varepsilon)} \right\rfloor, 0, g - 1 \right)$$

$\mathbf{u_f}$ is the integer cell coordinate; $\mathbf{g = 2^l}$ is grid width at octree level **l**; $\mathbf{b_{min}}$ is the mesh lower bound; **d** is mesh extent; and **epsilon** prevents division by zero on a flat axis.

u_f is the integer cell coordinate. Interleaved coordinate bits form a Morton code. Stable level and code sorting produce reproducible tile numbers. Centroid ownership preserves sum_i n_i = N, where N is the source count. Overlap ownership violates that identity and turns deduplication into a global topology problem.

4. Deterministic target allocation

For tile i, source count n_i, floor m_i = min(4,n_i), requested global target T, distributable capacity C, and remaining budget R:

$$C = \sum_i (n_i - m_i), \qquad R = T - \sum_i m_i$$

n_i is tile **i**'s source count; **m_i** is its minimum allocation; **C** is capacity above all floors; **T** is the requested global target; and **R** is the budget remaining after tile floors.

$$q_i = R \frac{n_i - m_i}{C}$$

q_i is tile **i**'s real-valued proportional quota; **R** is the remaining triangle budget; and **C** is total distributable capacity.

$$t_i = m_i + \lfloor q_i \rfloor + r_i, \qquad r_i \in \{0, 1\}, \qquad \sum_i t_i = T$$

t_i is the deterministic requested tile target; **floor(q_i)** is its whole-number quota; and **r_i** is the largest-remainder correction assigned with stable tile-order tie breaking.

q_i is the proportional real-valued quota. r_i is assigned in descending fractional-quota order, with tile number as the tie breaker. This largest-remainder construction reaches the requested sum before geometric constraints are applied.

If accepted tile results contain F'_i faces, the actual count and boundary overhead are:

$$T' = \sum_i |F'_i|, \qquad R_B = T' - T$$

F'_i is the accepted result-face set for tile **i**; **T'** is the actual assembled count; **T** is the request; and **R_B** is boundary-retention overhead in triangles.

R_B records geometry retained by the boundary contract. On the reference mesh, R_B = 495. A post-pass that deletes these faces would reach an exact counter value by bypassing the constraint that retained them.

5. Boundary invariant

Let c_i(e) be the number of tile-i triangles incident on edge e. The protected vertex set is:

$$B_i = \{v : v \in e \text{ and } c_i(e) = 1\}$$

$c_i(e)$ is edge **e**'s incident-face count inside tile **i**; **v** is an endpoint of an edge with one local incident face; and **B_i** is the protected tile-open vertex set.

Every protected source coordinate must appear in the optimized tile result at binary STL precision:

$$\forall v \in B_i, \ \exists v' \in V'_i : \ \text{float32}(\mathbf{x}_{v'}) = \text{float32}(\mathbf{x}_v)$$

V'_i is the optimized tile vertex set; **$\mathbf{x}_v$** and **$\mathbf{x}_{v'}$** are source and result coordinates; and **float32** applies binary STL precision. Every protected coordinate must survive bit-exactly.

B_i contains vertices on tile-open edges. V'_i is the optimized tile vertex set. Bit-exact float32 comparison matches STL storage and is stricter than a tolerance test.

6. Failure analysis

Attempt	Observed conflict	Resulting decision
Independent local QEM	Neighboring reducers can move the same seam differently	Freeze tile-open coordinates
Overlapping ownership	Duplicate geometry and ambiguous counts	One centroid owner per source face
Weld after assembly	Cannot reconstruct differently simplified boundary curves	Preserve before reduction
Force exact count	Deletes geometry retained for continuity	Report boundary overhead
Completion-order output	Binary result depends on scheduling	Assemble by stable tile number
CPU-count workers	Concurrent native state can exceed memory budget	Limit by memory and pending work
Trust existing files	Truncated or stale checkpoints can be reused	Validate structure, identity, and digest
Write final path directly	Cancellation can expose partial output	Validate temporary file, then replace atomically

For independently moved seam positions x'_L and x'_R, a crack exists when:

$$\|\mathbf{x}'_L - \mathbf{x}'_R\|_2 > 0$$

$\mathbf{x}'_L$ and $\mathbf{x}'_R$ are independently reduced positions for the same seam point; $\|\cdot\|_2$ is Euclidean distance. Any nonzero value is an open positional disagreement.

A welding tolerance can close some small gaps, but a large tolerance can merge intentional nearby surfaces. It also cannot restore a trajectory already simplified differently on each side.

7. Resource, checkpoint, and cancellation model

The persistent index stores raw 50-byte STL triangle records and limits simultaneously open tile streams. Each worker loads one tile, welds equal local coordinates, simplifies it, and atomically writes points, faces, metadata, and a checksum.

$$M_W = \max(64\,\mathrm{MiB},\ 320 n_{\max})$$

M_W is conservative memory per active worker; **n_{max}** is the largest pending tile count; 320 bytes per triangle estimates welded topology, quadrics, adjacency, candidates, and native temporary state.

$$W = \max\left(1, \min\left(W_{\mathrm{configured}}, \left\lfloor \frac{M_{\mathrm{budget}}}{M_W} \right\rfloor, N_P\right)\right)$$

W is active worker count; **$W_{configured}$** is the process ceiling; **M_{budget}** is available geometry memory; **M_W** is estimated memory per worker; and **N_P** is pending tile count.

M_W is the conservative per-worker estimate. n_max is the largest pending tile count. M_budget is the geometry-memory budget. N_P is pending work. W is active concurrency. The 320-byte factor covers welded topology, adjacency, quadrics, candidates, and native temporary state.

For checkpoint point bytes P_i and face bytes F_i:

$$H_i = \mathrm{SHA256}(P_i \| F_i)$$

P_i and **F_i** are the serialized point and face arrays for checkpoint **i**; **||** means byte concatenation; and **H_i** is the stored SHA-256 content digest.

Reuse also requires matching operation identity, valid array shapes, in-range face indices, and plausible face counts. Cancellation retains complete checkpoints but removes partial assembly output. A resumed run revalidates each tile instead of assuming that file existence means success.

GPU-guided reducers currently use one worker because independent processes can contend for the same selected device and repeat initialization. This is an explicit limitation pending matched tiled-device measurements.

8. Deterministic streamed assembly

For stable ascending tile order pi, assembly is:

$$F_{\mathrm{STL}} = F'_{\pi(1)} \,\|\, F'_{\pi(2)} \,\|\, \cdots \,\|\, F'_{\pi(k)}$$

pi is stable ascending tile order; $\mathbf{F'_{pi(i)}}$ is one validated tile-face sequence; **k** is tile count; and **||** means ordered record concatenation.

Worker completion order does not appear in the equation. Normals are recomputed for output vertices a, b, and c:

$$\mathbf{n} = \frac{(\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a})}{\|(\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a})\|_2}$$

a, **b**, and **c** are output triangle vertices; **x** is the cross product; $\mathbf{\|.\|_2}$ is Euclidean norm; and **n** is the recomputed unit normal. Degenerate triangles receive a zero normal.

A degenerate cross product receives a zero normal. Assembly loss and intentional reduction are different quantities:

$$L_{\mathrm{assembly}} = \sum_i |F'_i| - |F_{\mathrm{STL}}|$$

$\mathbf{L_{assembly}}$ is triangles lost during assembly; $\mathbf{|F'_i|}$ is an accepted tile count; and $\mathbf{|F_{STL}|}$ is the final written count. A correct assembly has zero loss.

$$R_S = 1 - \frac{|F_{\mathrm{STL}}|}{|F_{\mathrm{source}}|}$$

$\mathbf{R_S}$ is the intended simplification fraction; $\mathbf{|F_{STL}|}$ is result count; and $\mathbf{|F_{source}|}$ is source count. This is separate from assembly loss.

L_assembly measures whether validated output disappeared during writing. R_S measures intended simplification. All measured runs had L_assembly = 0.

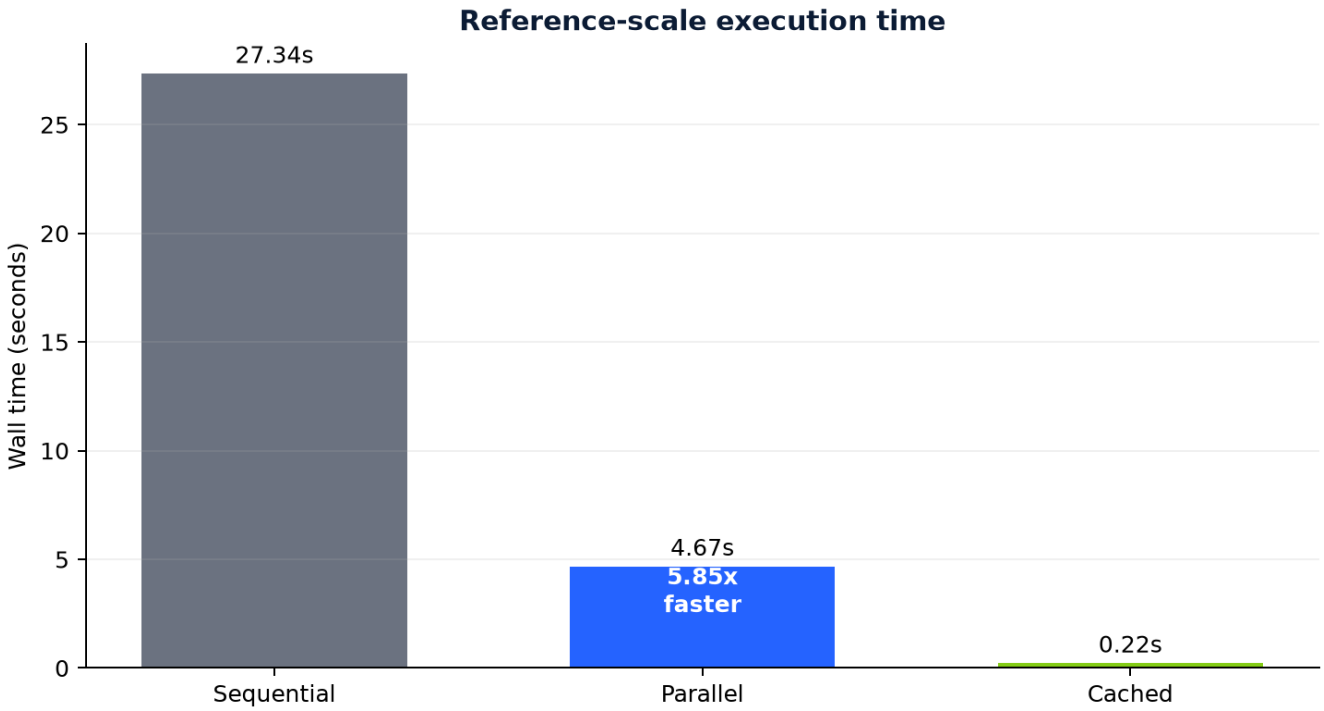
Source fingerprint H_s, target T, algorithm A, and settings G determine the output hash under a deterministic local reducer:

$$(H_s, T, A, G) \longmapsto H_{\mathrm{out}}$$

$\mathbf{H_s}$ is source identity; **T** is requested target; **A** is algorithm identity; **G** is the complete settings set; and $\mathbf{H_{out}}$ is the final output hash.

Stable spatial order, stable quota tie breaking, validated tile identity, and ordered concatenation make scheduling affect runtime without affecting H_out.

9. Performance



Run	Workers	Time	Cached tiles	Output triangles	Assembly loss
Sequential	1	27.336854 s	0	249,679	0
Parallel	8	4.671751 s	0	249,679	0
Cached validation and assembly	1 active	0.215439 s	50	249,679	0

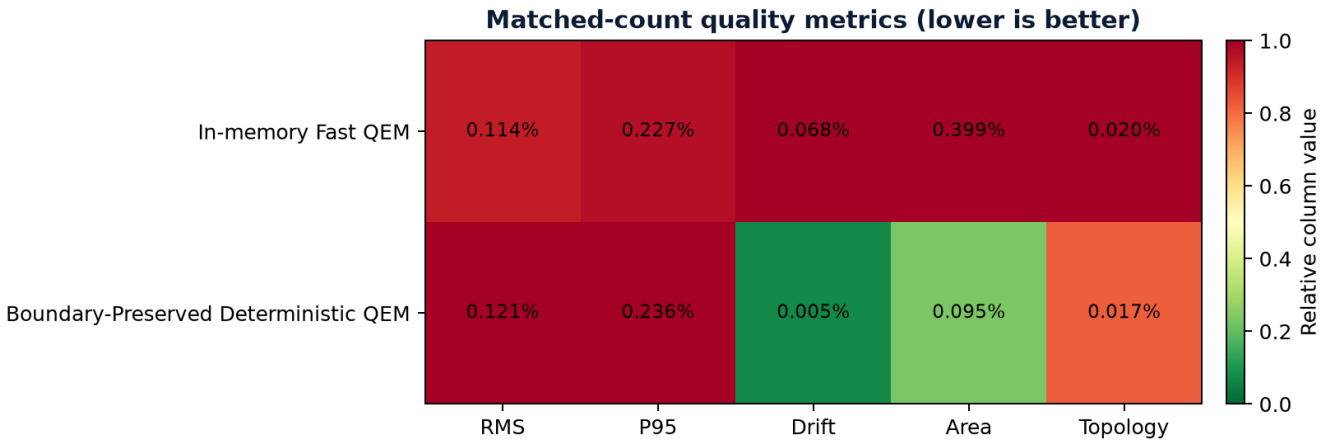
$$S_W = \frac{t_1}{t_W}, \qquad \eta_W = \frac{S_W}{W}$$

t_1 is one-worker time; **t_W** is time with **W** workers; **S_W** is speedup; and **eta_W** is parallel efficiency on a 0-to-1 scale.

S_W is speedup and eta_W is worker efficiency. The measured S_8 was 5.85 and eta_8 was 0.73. Process startup, storage traffic, unequal tile topology, serialization, and final assembly account for the difference from ideal scaling. The cached trial measures validation and reconstruction, not another simplification pass.

All outputs were 12,484,034 bytes with SHA-256 00455d33955b99149db58e70293484cacbc71372879ea61bcd693abd2133c713.

10. Quality measurements



Measurement	In-memory Fast QEM	Boundary-Preserved Deterministic QEM
Runtime	8.581000 s	4.671751 s
Output triangles	249,184	249,679
RMS sampled surface error	0.113647%	0.120825%
P95 sampled surface error	0.227477%	0.235518%
Dimension drift	0.067879%	0.004893%
Surface-area change	0.399084%	0.094952%
Non-manifold rate	0.020467%	0.016855%

11. Quality formulas

For sampled source point x_s, nearest output point y_s, source diagonal D, and sample count S:

$$D_{\mathrm{RMS}} = \frac{100}{D}\sqrt{\frac{1}{S}\sum_{s=1}^{S}\|\mathbf{x}_s - \mathbf{y}_s\|_2^2}$$

$\mathbf{x}_s$ is sampled source point **s**; $\mathbf{y}_s$ is its nearest result point; **S** is sample count; **D** is source diagonal; and $\mathbf{D}_{\mathbf{RMS}}$ is average normalized surface displacement in percent.

$$D_{95} = \frac{100}{D}\,\mathrm{percentile}_{95}(\|\mathbf{x}_s - \mathbf{y}_s\|_2)$$

$\mathbf{D}_{\mathbf{95}}$ is the normalized 95th-percentile sampled displacement; $\mathbf{x}_{\mathbf{s}}$, $\mathbf{y}_{\mathbf{s}}$, and **D** have the same meanings as above. It exposes the higher-error tail that an average can hide.

D_RMS measures average sampled displacement. D_95 exposes the higher-error tail. Normalizing by D makes the result comparable across model scales.

For source extent d_j, output extent d'_j, source area A, output area A', output edge incidence c'(e), and output faces F':

$$\delta_j = 100\frac{|d'_j - d_j|}{D}$$

$\mathbf{d}_{\mathbf{j}}$ and $\mathbf{d}'_{\mathbf{j}}$ are source and result extents on axis **j**; **D** is source diagonal; and $\mathbf{delta}_{\mathbf{j}}$ is normalized dimension drift in percent.

$$D_A = 100\frac{|A' - A|}{A}$$

A and $\mathbf{A}'$ are source and result surface areas; $\mathbf{D}_{\mathbf{A}}$ is absolute relative area change in percent.

$$D_N = 100\frac{|\{e : c'(e) > 2\}|}{|F'|}$$

$\mathbf{c}'\mathbf{(e)}$ is result edge incidence; edges above two are non-manifold; $\mathbf{|F'|}$ is result triangle count; and $\mathbf{D}_{\mathbf{N}}$ is the normalized non-manifold rate in percent.

delta_j measures bounding-dimension drift on axis j. D_A measures surface-area change. D_N measures non-manifold edge incidence relative to result size. No single value proves quality. RMS can hide isolated damage, P95 can miss rare extremes, bounds can remain stable while interior detail changes, and topology can improve while distance rises slightly. The metrics are interpreted together.

On the reference mesh, the tiled result traded 0.007178 percentage points of RMS and 0.008041 points of P95 error for lower drift, area change, non-manifold rate, and runtime. These values characterize this fixture and machine, not every mesh class.

12. Validation contracts

Validated production contracts

Contract	Result	Evidence
Source Partition	PASS	sum of tile source counts equals 4126315
Deterministic Target	PASS	repeated largest-remainder allocation is identical and sums to requested target
Assembly Count	PASS	assembled record count equals 249679 validated tile triangles
Assembly Payload	PASS	assembled triangle payload equals ordered tile payload concatenation
Parallel Consistency	PASS	sequential and parallel SHA-256 values match
Checkpoint Consistency	PASS	cached reconstruction SHA-256 matches fresh runs
Connected Seams	PASS	source and output boundary-edge counts match
Cancellation	PASS	no published STL or partial file remains
Resume	PASS	completed checkpoint reused and final hash matches
Corruption	PASS	invalid checkpoint rejected and rebuilt

A 260 by 130 quad grid produces 67,600 triangles and crosses several tiles. With exact-coordinate welding, boundary-edge count is:

$$N_B = |\{e : c(e) = 1\}|$$

c(e) is exact-coordinate welded edge incidence; **e** is an undirected edge; and $\mathbf{N_B}$ is the number of boundary edges. Matching source and output values detects newly opened seams on the connected fixture.

The contract requires N_B_output = N_B_source. This detects newly opened seams on the connected fixture. It does not claim to repair openings already present in the source.

Contract	Observed result
Source ownership	Every source triangle assigned once
Assembly count	Equals sum of validated tile outputs
Artificial open seams	0 detected
Sequential versus parallel	Byte-identical
Fresh versus cached	Byte-identical
Cancellation	No final or partial STL published
Resume	Completed checkpoints reused
Corrupt checkpoint	Rejected and rebuilt

13. Evidence limits and future work

The evidence establishes deterministic bytes across tested schedules, zero assembly loss, cancellation cleanup, resumable validated checkpoints, and no new open edges on the connected fixture. It does not prove watertightness, manifoldness, absence of self-intersections, or universal quality for arbitrary inputs.

Boundary retention is conservative. A future shared-boundary optimizer could reduce R_B, but it needs the same deterministic identity, cancellation, and recovery contracts. Additional adversarial fixtures should include curved seams, thin shells, intersecting layers, pre-existing non-manifold edges, disconnected components, degenerate triangles, and extreme coordinate ranges.

The current index targets binary STL. Formats with shared normals, colors, materials, or texture coordinates require streamed readers and explicit attribute-preservation rules. A complete source hash can supplement the bounded fast invalidation fingerprint when cryptographic source identity is required.

14. Prior work and attribution

Garland and Heckbert established iterative pair contraction with accumulated quadric error matrices. Lindstrom established out-of-core simplification using clustering and quadric placement. Cignoni and colleagues described external-memory octree processing with protected block-boundary vertices. Their work supplied essential mathematical and systems foundations. This report focuses on the production integration and the measured contracts around those foundations.

1. M. Garland and P. Heckbert. Surface Simplification Using Quadric Error Metrics. SIGGRAPH 1997.
2. P. Lindstrom. Out-of-Core Simplification of Large Polygonal Models. SIGGRAPH 2000.
3. P. Cignoni et al. External Memory Management and Simplification of Huge Meshes.

15. Reproduction

The publication package includes a frozen implementation snapshot, contract tests, benchmark driver, normalized CSV and JSON evidence, figure generator, experiment catalog, attribution record, and detailed algorithm and design-decision notes. The private reference mesh is not distributed.

```
python benchmark_boundary_preserved_deterministic_qem.py INPUT.stl --target 249184 --workers 8
--output OUTPUT_DIRECTORY
```

The benchmark clears result state between fresh sequential and parallel runs, retains checkpoints for the cached run, emits machine-readable results, and fails if output hashes or assembled counts disagree.